MILITARY SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, CIRCULAR THREADED, HIGH SHOCK, HIGH DENSITY, SHIPBOARD, BACKSHELL, STRAIGHT, CABLE SEALING AND SHIELD TERMINATION, CONNECTOP ELECTRIC

This specification sheet is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the accessories described herein shall consist of this document and the latest issue of specification MIL-C-28840.

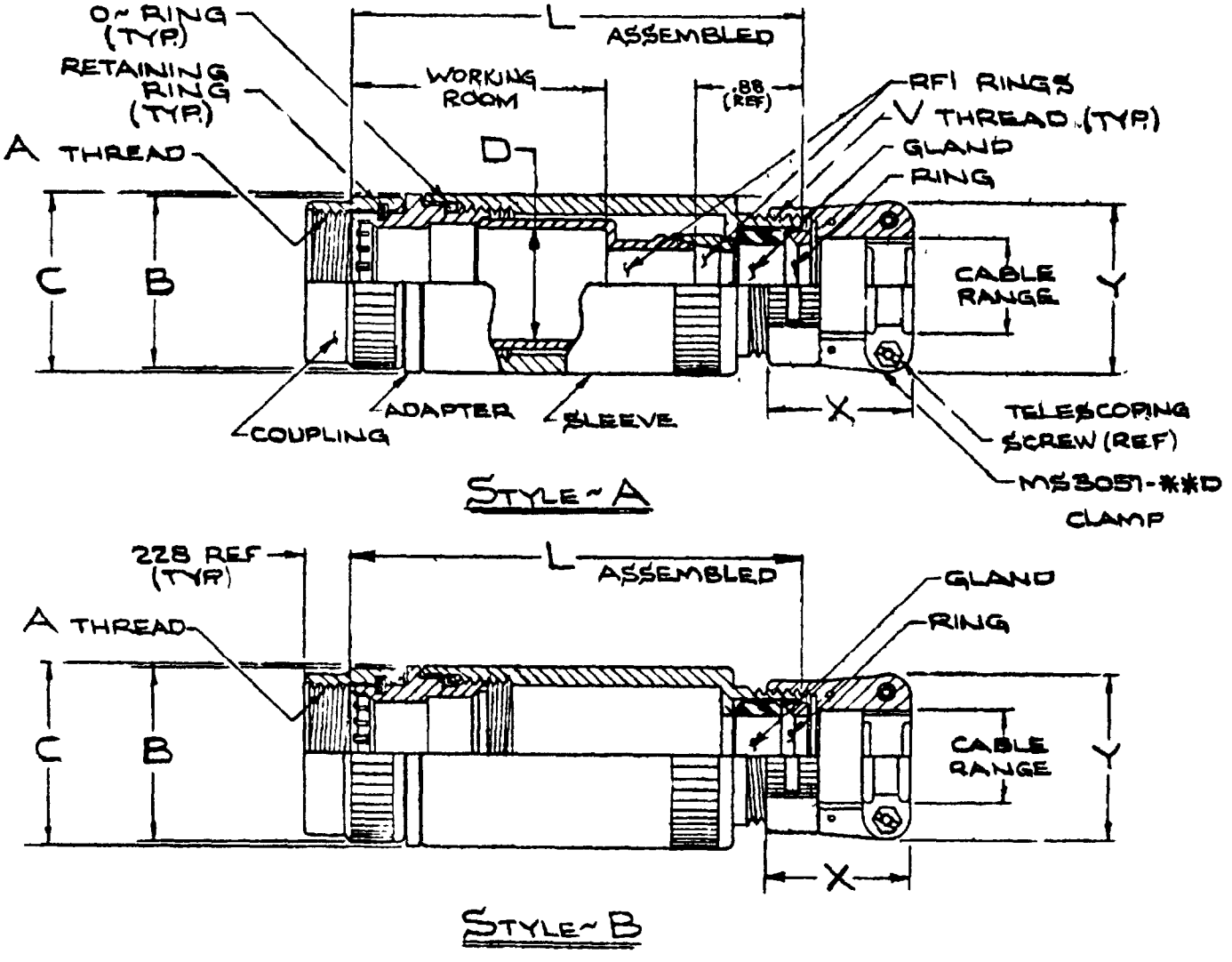


FIGURE 1. Dimensions and configurations

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:33Z
Check the source to verify that this is the current version before use.

TABLE I Dimensions and configurations.

DASH NO.	DESIGNATOR 1/	A THREAD CLASS 2B	B MAX. DIA.	C MAX. DIA.	D MIN DIA.	CABLE RANGE MAX.	CABLE RANGE MIN.	L ±.12 (±3.05)	WORKING ROOM REF.	V THREAD CLASS 2	X REF DIM.	Y REF. DIM.
01	A(11)	.750-20 UNEF	1.028 (26.11)	1.156 (29.36)	.750 (19.05)	.312 (7.92)	.188 (4.77)	3.62 (91.95)	2.00 (50.80)	.625-24 UNEF	1.281 (32.53)	.938 (23.82)
02	B(13)	.875-20 UNEF	1.141 (28.98)	1.156 (29.36)	.750 (19.05	.312 (7.92)	.188 (4.77)			.625-24 UNEF	1.281 (32.54)	.938 (23.82)
03						.438 (11.12)	.281 (7.14)			.750-20 UNEF	1.281 (32.54)	1.125 (28.57)
04	C(15)	1.00-20 UNEF	1.263 (32.08)	1.281 (32.54)	.875 (22.23)	.312 (7.92)	.188 (4.77)			.625-24 UNEF	1.281 (32.54)	.938 (23.82)
05						.438 (11.12)	.281 (7.14)			.750-20 UNEF	1.281 (32.54)	1.125 (28.57)
06						.562 (14.27)	.344 (8.74)			.875-20 UNEF	1.281 (32.54)	1.312 (33.32)
07				1.406 (35.71)	1.000 (25.40)	.750 (19.05)	.438 (11.12)			1.188-18 UNEF	1.312 (33.32)	1.593 (40.62)
08	D(17)	1.125-18 UNEF	1.387 (35.23)			.625 (15.87)	.375 (9.52)			1.000-20 UNEF	1.281 (32.54)	1.312 (33.32)
09						.750 (19.05)	.438 (11.12)	4.12 (104.65)	2.50 (63.50)	1.188-18 UNEF	1.312 (33.32)	1.593 (40.62)
10				1.531 (38.89)	1.125 (28.58)	938 23.82)	.625 (15.87)			1.438-18 UNEF	1.406 (35.71)	1.750 (44.45)
11	E(19)	1.250-18 UNEF	1.513 (38.43)			.625 (15.87)	.375 (9.52)			1.000-20 UNEF	1.281 (32.54)	1.312 (33.32)
12						.938 (23.82)	.625 (15.87)			1.438-18 UNEF	1.406 (35.71)	1.750 (44.45)
13	F(23)	1.438-18 UNEF	1.703 (43.51)	1.656 (42.06)	1.250 (31.75)	.750 (19.05)	.438 (11.12)			1.188-18 UNEF	1.312 (33.32)	1.593 (40.62)
14						.938 (23.82)	.625 (15.87)	4.62 (117.35)	3.00 (76.20)	1.438-18 UNEF	1.406 (35.71)	1.750 (44.45)
15				1.781 (45.24)	1.375 (34.93)	1.188 (30.17)	.812 (20.62)			1.750-18 UNS	1.593 (40.46)	2.093 (53.16)
16	G(25)	1.562-18 UNEF	1.825 (46.35)			.938 (23.82)	.625 (15.87)			1.438-18 UNEF	1.406 (35.71)	1.750 (44.45)
17						1.188 (30.17)	.812 (20.62)			1.750-18 UNS	1.593 (40.46)	2.093 (53.16)
18						1.250 (31.75)	.875 (22.23)			1.750-18UNS	1.593 (40.46)	2.093 (53.16)
19	H(29)	1.875-16 UN	2.143 (54.43)	2.156 (54.76)	1.750 (44.45)	1.386 (35.05)	1.000 (25.40)	5.12 (130.05)	3.50 (88.90)	2.000-18 UNS	1.625 (41.28)	2.343 (59.51)
20						1.188 (30.17)	.812 (20.62)			1.750-18 UNS	1.593 (40.46)	2.093 (53.16)
21						1.380 (38.05)	1.000 (25.40)			2.000-18 UNS	1.625 (41.28)	2.343 (59.51)
22	I(33)	2.062-16 UNS	2.329 (59.16)	2.281 (57.94)	1.875 (47.63)	1.350 (35.05)	1.000 (25.40)			2.000-18 UNS	1.625 (41.28)	2.343 (59.51)
23						1.625 (41.28)	1.250 (31.75)			2.250-16 UNS	1.900 (48.26)	2.750 (69.85)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:33Z
Check the source to verify that this is the current version before use.

TABLE I. Dimensions and configurations(continued).

DASH NO.	DESIGNATOR 1/	A THREAD CLASS 2B	B MAX. DIA.	C MAX. DIA.	D MIN DIA.	CABLE RANGE MAX.	CABLE RANGE MIN.	L ±.12 (±3.05)	WORKING ROOM REF.	V THREAD CLASS 2	X REF DIM.	Y REF. DIM.
24	A(11)	.750-20 UNEF	1.028 (26.11)	1.156 (29.36)	.750 (19.05)	.438 (11.12)	.281 (7.14)	3.62 (91.95)	2.00 (50.80)	.750-20 UNEF	1.281 (32.54)	1.125 (28.57)
25				1.281 (32.54)	.875 (22.22)	.562 (14.27)	.344 (8.74)	3.62 (91.95)	2.00 (50.80)	.875-20 UNEF	1.281 (32.54)	1.312 (33.32)
26	B(13)	.875-20 UNEF	1.141 (28.98)	1.281 (32.54)	.875 (22.22)	.562 (14.27)	.344 (8.74)	3.62 (91.95)	2.00 (50.80)	.875-20 UNEF	1.281 (32.54)	1.312 (33.32)
27				1.406 (35.71)	1.000 (25.4)	.625 (15.87)	.375 (9.52)	4.12 (104.65)	2.50 (63.50)	1.000-20 UNEF	1.281 (32.54)	1.312 (33.32)
28	C(15)	1.000-20 UNEF	1.263 (32.08)			.625 (15.87)	.375 (9.52)			1.000-20 UNEF	1.281 (32.54)	1.312 (33.32)

NOTES

1. Shell sizes are provided within parentheses for information and are not a part of the designator.
2. Dimensions are in inches.
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1.0 inch equals 25.4 mm.
4. Dimensions apply after plating.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:33Z
Check the source to verify that this is the current version before use.

REQUIREMENTS

Dimensions and configurations. See FIGURE 1 and TABLE I. Interface dimensions shall conform to MIL-C-28840, figure 7.

Material shall be aluminum alloy, in accordance with QQ-A-225, QQ-A-591, or QQ-A-601; or corrosion resistant steel, in accordance with QQ-S-763, class 316.

Finish: Aluminum components shall be cadmium plate, olive drab over electroless nickel.
Corrosion resistant steel components shall be cadmium plate, black. The screws and washers shall be passivated in accordance with QQ-P-35.

Marking: The backshell shall be identified with the manufacturer's name or trade-mark and the part number. The example of the part number is as follows.

M28840 /6 16 W A

MILITARY DESIGNATOR

SPECIFICATIONS SHEET NUMBER

DASH NUMBER

STYLE DESIGNATOR

A-EMI/ENVIRONMENTAL
B-ENVIRONMENTAL

MATERIAL AND FINISH DESIGNATOR

W-ALUMINUM, CADMIUM PLATE, OLIVE DRAB OVER ELECTROLESS NICKEL
D-CORROSION RESISTANT STEEL CADMIUM PLATE, BLACK

QUALIFICATION. Two samples in each shell size for which qualification is desired shall be tested in accordance with MIL-C-28840, TABLE V, Group 5.

Revision letters are not used to denote changes due to the extensiveness of the changes.

User activities.
Navy - OS, SH

Preparing activity
Navy-EC
(Project 5935-N219)

U.S. GOVERNMENT PRINTING OFFICE: 1980 - 605-631: 218
Source: https://assist.dla.mil - Downloaded: 2026-10-20T19:33Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY

Naval Electronic Systems Command
Washington, D. C. 20363

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

BUSINESS REPLY MAIL

FIRST CLASS PERMIT NO 12503 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

COMMANDER
NAVAL ELECTRONIC SYSTEMS COMMAND
DEFENSE STANDARDIZATION PROGRAM BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20363

ATTN: ELEX 8111

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:33Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS This form is provided to solicit beneficial comments which may improve this document and enhance its use DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government Fold on lines on reverse side, staple in corner, and send to preparing activity Attach any pertinent data which may be of use in improving this document If there are additional papers, attach to form and place both in an envelope addressed to preparing activity A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed

NOTE This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements

DOCUMENT IDENTIFIER (Number) AND TITLE

MIL-C-28840/6B(EC)

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1 ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDED WORDING CHANGE

C REASON FOR RECOMMENDED CHANGE(S)

2 REMARKS

SUBMITTED BY (Printed or typed name and address — Optional)

TELEPHONE NO

DATE

DD FORM 1 OCT 76 1426 EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:33Z
Check the source to verify that this is the current version before use.